

AptitudeMind

Updated Complete Development Record & Roadmap

An Adaptive Agentic AI Mentor

This document updates the previous AptitudeMind Complete Development Record. It preserves the original vision and records the newer Question Adapter, Profit Percentage verification, end-to-end Agent testing, and current remaining work.

Latest verified milestone: Question Bank → Question Adapter → Validator → Evaluator → Answer Engine → Correct Result.

1. Project Identity

AptitudeMind is a local-first adaptive aptitude mentor designed to understand topic, difficulty and learner performance, retrieve or generate questions, verify them deterministically, evaluate answers, store progress and adapt future practice.

2. Original Full Vision — Preserved

- Quantitative Aptitude, Logical Reasoning and Verbal Ability
- AI-generated and verified questions
- Adaptive difficulty and learner performance memory
- Weak-topic analysis
- RAG / aptitude knowledge retrieval
- Agent decision-making and multi-step workflows
- Validation, guardrails and self-correction
- Testing and evaluation
- Backend/API
- Student dashboard and practice arena
- AI mentor chat
- Topic library, analytics and progress
- Gamification
- Deployment
- Final documentation, demo and presentation

3. Development Environment

Item	Current setup
OS	Ubuntu Linux
Editor	VS Code
Python	3.12.3
LLM runtime	Ollama 0.33.3
Local model	llama3.2:3b
Project	~/AptitudeMind
Environment	.venv
Version control	Git / GitHub

4. Architecture and Adaptive Flow

Student → Agent → Retrieve/Generate → Adapt → Validate → Evaluate → Memory → Adaptive next question

The new Question Adapter sits between legacy Question Bank records and the standardized evaluation pipeline.

Question Bank → **Question Adapter** → Standard Question → Validator/Evaluator → Answer Engine → Result

5. Earlier Development Record — Status Snapshot

Area	Earlier recorded status
Ollama / local LLM	100%
Calculator / tools	100%
Persistent memory	100%
Topic system	~90%
Company system	~90%
Question bank / search	~85%
Adaptive difficulty	~90%
Agent decision layer	~85%
Agent execution layer	~80%
Validator / guardrails	~70%
Self-correction / regeneration	~70%
RAG	~0%
Complete agent workflow	~50%
Website / UI	~0%
API / backend	~0%
Testing / evaluation	~30%
Deployment	~0%
Final documentation / demo	~40%

The previous document estimated overall completion toward the long-term vision at approximately 65%. This was a development estimate, not a formal engineering metric. No new overall percentage is invented here.

6. New Work Completed After the Previous PDF

- ✓ **Question Adapter** — Created question_adapter.py to convert legacy Question Bank records into the standardized evaluator format.
- ✓ **Profit Percentage support** — Added deterministic Profit Percentage calculation to the Answer Engine.
- ✓ **Profit and Loss routing** — Integrated Profit and Loss / PROFIT_PERCENTAGE routing so adapted questions reach the deterministic solver.
- ✓ **Adapter test** — Passed: ■500 cost price, ■600 selling price, four options generated.
- ✓ **Answer Engine tests** — Reached 15/15 passing tests, including direct and adapted Profit Percentage tests.
- ✓ **Agent integration** — Retrieved Infosys questions now pass through adaptation and retrieved-question validation.
- ✓ **End-to-end test** — Infosys retrieval → adaptation → validation → student answer → calculation → evaluation succeeded.
- ✓ **Verified result** — Student option 3 was evaluated as correct; calculated answer was 20.0%.
- ✓ **Git milestone** — agent.py, answer_engine.py and question_adapter.py were staged and committed locally; working tree became clean.

7. Verified End-to-End Result

Request: Infosys

Question: An article is bought for ■500 and sold for ■600. What is the profit percentage?

Options: 10, 15, 20, 25

Student answer: option 3

Calculated answer: 20.0%

Evaluation: **CORRECT**

This confirms that the adapter bridges the older Question Bank format with the newer deterministic evaluation pipeline.

8. Current Core Pipeline

Stage	Current state
User request	Working
Search / filters	Working
Question Bank retrieval	Working
Question Adapter	Working for PROFIT_PERCENTAGE
Question validation	Working
Answer Engine	Working for supported deterministic types
Student evaluation	Working
Progress / memory	Existing; broader integration still needs verification
Adaptive next question	Partially implemented; broader workflow remains

9. Remaining Work — Not Yet Completed

- **Finish deterministic coverage** — Expand adapter and Answer Engine coverage across the existing question-bank types.
- **Stabilize complete agent workflow** — Verify retrieve/generate → adapt → validate → regenerate → evaluate → memory → adaptive next-question flow.
- **RAG / knowledge retrieval** — Build the aptitude knowledge base, retrieval layer and grounded mentor explanations.
- **Testing / evaluation** — Add broader unit, integration, regression, edge-case and generated-question tests.
- **Backend / API** — Expose core AptitudeMind capabilities through a clean application API.
- **Website / UI** — Build dashboard, practice arena, topic selection, question interface, results, progress and mentor chat.
- **Analytics** — Complete weak-topic visualization and learner analytics.
- **Deployment** — Package and deploy the usable application.
- **Final documentation** — Update README, architecture documentation, screenshots, setup instructions and testing evidence.
- **Final presentation** — Prepare a reliable end-to-end demonstration.

10. Recommended Development Order

1. Finish Question Adapter + Answer Engine coverage
2. Run comprehensive evaluator and agent integration tests
3. Stabilize the complete adaptive agent workflow
4. Build RAG / aptitude knowledge retrieval
5. Build backend/API
6. Build website/dashboard/practice arena
7. Add analytics and AI mentor chat
8. Test the complete application
9. Deploy
10. Final documentation, demo and presentation

11. Engineering Principles

- Use deterministic Python verification for mathematical correctness.

- Treat the LLM as a generator/reasoning component, not an unquestioned source of truth.
- Validate generated and retrieved questions before presenting them.
- Persist learner history for adaptation.
- Build → test → debug → connect.
- Preserve the full project vision.
- Preserve successful implementations, bugs, debugging decisions and fixes.
- Understand the architecture instead of blindly copying an agent.

12. Current Checkpoint

The project has moved beyond the earlier validator-focused checkpoint. The latest verified milestone is successful integration of the Question Adapter with the Agent, Evaluator and deterministic Answer Engine. The next priority is to generalize this integration across remaining question types and stabilize the complete adaptive workflow before moving into RAG, API and UI layers.

Repository note: at the time of this record, the local milestone commit existed and the working tree was clean; the final push to origin/main remained the repository synchronization step.